

LET'S LEARN PYTHON

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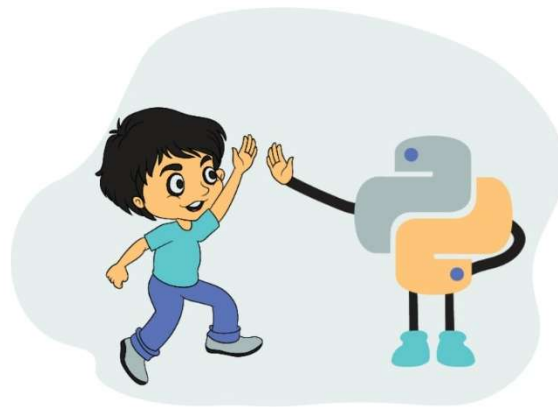
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Introduction

What is Python?

Created by Guido van Rossum, Python is an object-oriented, general-purpose, high-level programming language.

The main aim of Python was to make programming easier for beginners and allow developers to get things done in fewer lines of code compared to other programming languages. Because of this simplicity, Python has become one of the most popular choices for learning programming in recent times.



Python is:

- General-purpose → can be used in web development, data science, AI, automation, and more.
- Interpreted → code runs line by line, no need for compilation.
- Interactive → you can test code directly in its shell (REPL).
- Object-Oriented → supports classes and objects for structured programming.
- Readable → uses English keywords and simple syntax rather than complex symbols.

Python is a perfect choice of programming language if you are a beginner. If you have any previous programming experience, then learning Python would be a cakewalk.

History of Python:

- 1985–1990: Guido van Rossum started working on Python at CWI (Centrum Wiskunde & Informatica) in the Netherlands.
- Inspiration: Python was designed as a successor to the ABC programming language. ABC had good features like exception handling and worked with the Amoeba operating system, but it also had some limitations.



- Guido wanted to keep the good parts of ABC while fixing its flaws. This led him to create a new scripting language that was simple, powerful, and practical.
- 1991: Python was officially released. The first version already included important features such as functions, exception handling, and core data types like strings, lists, and dictionaries.
- The Name “Python”:

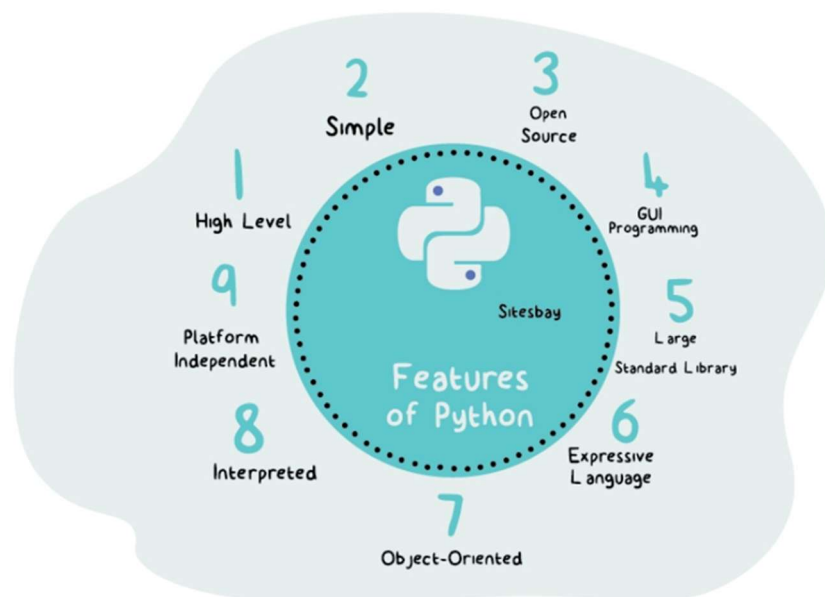
The name did not come from the snake 🐍. Guido was a big fan of the BBC comedy show “Monty Python’s Flying Circus”. He wanted a name that was short, unique, and slightly mysterious, so he chose Python.



And that’s how one of the world’s most loved programming languages was born. Amazing, right?

Features of Python:

- Easy-to-learn - Python has few keywords, simple structure, and a clearly defined syntax. This makes it easy to catch for beginners.
- Easy-to-read - Python code is more clearly defined and visible to the eyes.
- Easy-to-maintain - Python's source code is fairly easy-to-maintain
- Broad standard library - Python's bulk libraries are very portable and cross-platform compatible with UNIX, Windows, and Macintosh.
- Interactive Mode - Python has support for an interactive mode that allows interactive testing and debugging of snippets of code.
- Portable - Python can run on a wide variety of hardware platforms and has the same interface on all of those.
- Extendable - You can add low-level modules to the Python interpreter. These modules enable programmers to add or customize their tools to be more efficient



Why Learn Python?

Do you remember that Python is a general-purpose programming language?

That means it can be used almost everywhere — and this is exactly why developers around the world love Python so much.

Python is used for:

- Web Development
- App Development
- Data Science
- Machine Learning
- Internet of Things (IoT)

...and the list goes on!

Let's deep-dive into some of these areas:

Web Development

Python is one of the top choices for backend web development (not frontend).

Django → A powerful framework used for building large, full-fledged applications.

Flask → A lightweight framework suitable for smaller applications.

Both frameworks are in high demand, and developers skilled in Django or Flask often get high-paying roles.



App Development

Python can also be used for mobile applications.

While not the most popular choice for mobile apps,

Python has a framework called Kivy, which allows you to build cross-platform apps (apps that run on both Android and iOS).

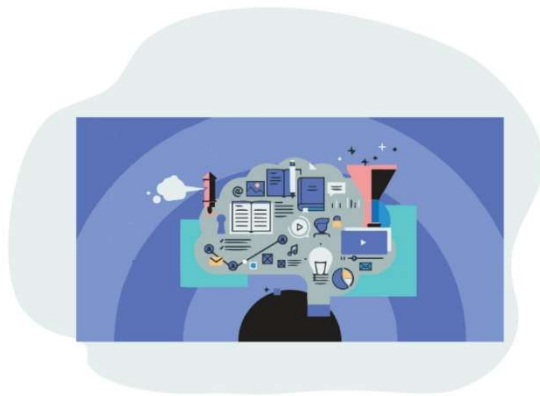
Data Science

One of the biggest reasons for Python's popularity today is its dominance in Data Science.

Python offers libraries for data processing, analysis, and visualization.

It has become the go-to language for data scientists because of its simplicity + powerful tools.

Skills in Python + Data Science are in very high demand in today's industry.



Machine Learning

Machine Learning might sound complicated, but Python makes it accessible even for beginners. With libraries like Keras and Scikit-learn, you can build powerful ML models without diving deep into complex math and algorithms.

This makes Python a top choice for anyone who wants to step into AI and ML.

Interpreted Python & Its Versions:

Of course, we know that Python is a programming language. But did you know there are two main types of programming languages?

1. Compiled Language
2. Interpreted Language

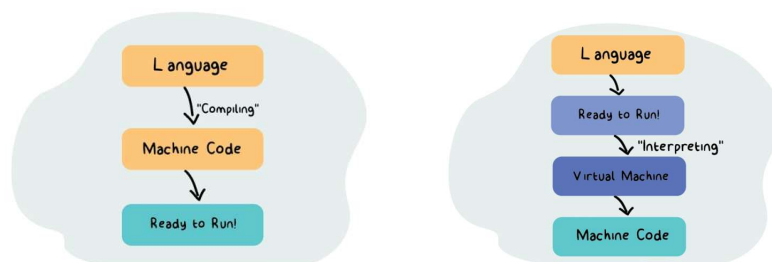
Chances of Error

- Compiled Languages:

In a compiled language, your program is first converted into machine code before it runs. If there are any errors in your code, the compiler won't let the program execute until all errors are fixed. This means you see all errors at once before running the program.

- Interpreted Languages:

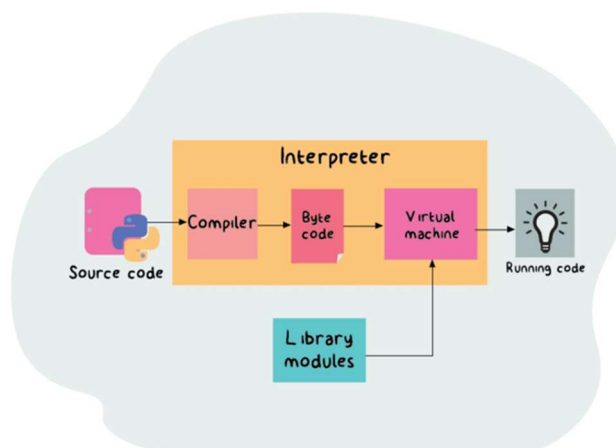
An interpreted language starts running the code immediately, even if there are errors in parts of the program. When an error is encountered, the program stops execution, and an error message shows the problem. This allows you to test code line by line.

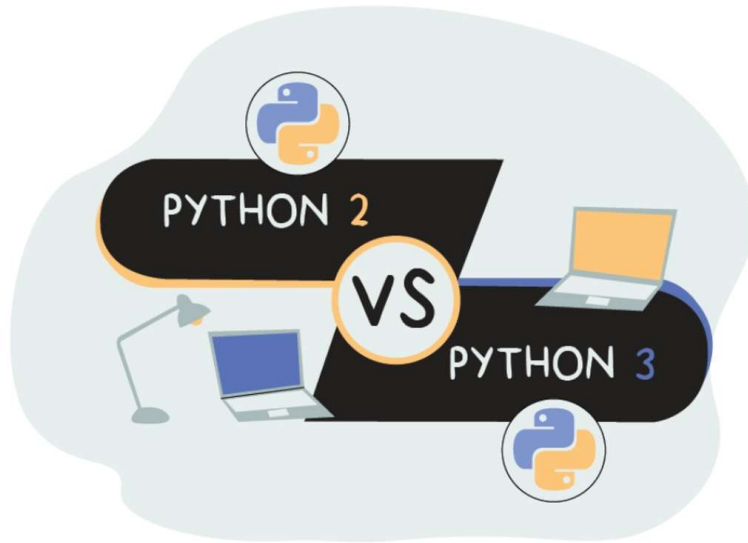


Python is an Interpreted Language

Python is a classic example of an interpreted programming language.

- Most Python implementations execute instructions directly without compiling the entire program into machine code.
- The interpreter translates each statement into subroutines and then into machine code as the program runs.
- This makes Python flexible, interactive, and beginner-friendly, because you can write and test code quickly.





Python Versions

- **Python 2 → Legacy Version**
 - Python 2 was popular for over a decade, and some companies still use it.
 - However, it is slowly becoming obsolete.
- **Python 3 → Current and Future Version**
 - Python 3 fixes many of Python 2's limitations and is actively supported.
 - For beginners, it is recommended to learn Python 3, as most modern development uses it.

Installing Python & Setting up IDEs

PYTHON BASICS